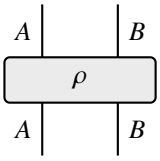
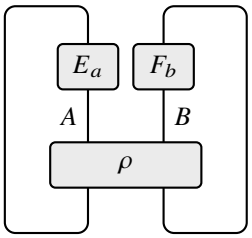
Process matrices in pictures (operator convention)

Every box is an operator. Its bra indices are wires entering from below, its ket indices are wires leaving above. Placing one box above another and joining wires is operator multiplication; a wire looped from the top of a diagram back to its bottom is a trace. A bare vertical wire is the identity. A dashed red line is a horizontal cut: the rank of what is drawn is at most the product of the dimensions of the wires the cut crosses.

1. A state is an operator: the ordinary Born rule



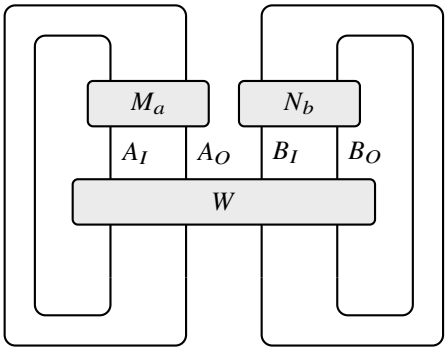
ρ: two wires in, two wires out



$= \mathrm{Tr}\big[\rho\,(E_a \otimes F_b)\big]$

closing every wire gives a number

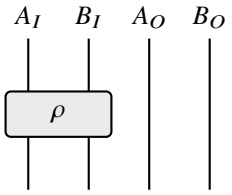
2. The process matrix W and its Born rule



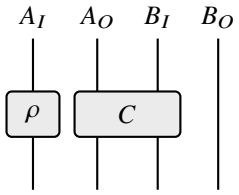
$p(a,b) = \mathrm{Tr}\big[W\,(M_a \otimes N_b)\big]$

Alice's instrument element is a CP map $A_I \to A_O$, but we never draw the map: we draw its Choi matrix M_a, an operator on $A_I \otimes A_O$, so a box with two wires in and two wires out, exactly like ρ in picture 1. Likewise N_b. The process matrix W is an operator on all four wires. The Born rule is the same closed diagram as in picture 1 with $\rho \to W$ and $E_a, F_b \to M_a, N_b$. Validity of W: the closed diagram equals 1 whenever M_a, N_b are summed over outcomes. There is no separate ρ in this formalism; W is the only state-like object, and any state handed to the laboratories is part of W (next picture).

3. Two familiar W's, drawn as operators



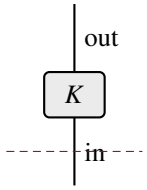
(a) $W = \rho_{A_I B_I} \otimes 1_{A_O} \otimes 1_{B_O}$
(wires in the order $A_I B_I \mid A_O B_O$)



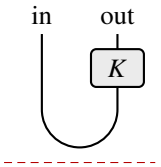
(b) $W = \rho_{A_I} \otimes C_{A_O B_I} \otimes 1_{B_O}$
Alice first

In (a), contracting with $M_a \otimes N_b$ as in picture 2, the bare wires A_O, B_O just trace M_a and N_b over their output index, leaving POVM elements on A_I, B_I, and one recovers picture 1. In (b), C is the Choi matrix of a channel: the box sits on the pair (A_O, B_I) because it connects Alice's output to Bob's input. A causally separable W is a convex combination of (b) and its mirror image. The content of the subject is that there are valid W's of neither form.

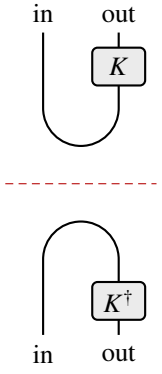
4. The Choi isomorphism, and what "rank one" looks like



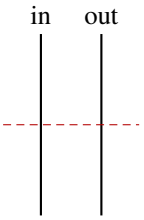
(a) a map K
cut crosses one wire
rank $\leq \dim(\text{in})$



(b) $|K\rangle\rangle = (1 \otimes K)|I\rangle\rangle$
the "in" wire is bent up
cut crosses nothing: a vector



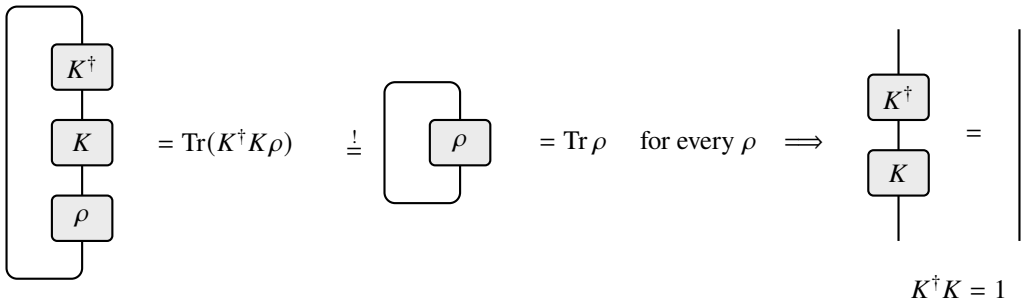
(c) $J(K) = |K\rangle\rangle\langle\langle K|$
two wires in, two out, but the
cut crosses nothing: **rank one**



(d) $1_{\text{in}\otimes\text{out}}$
cut crosses two wires
full rank

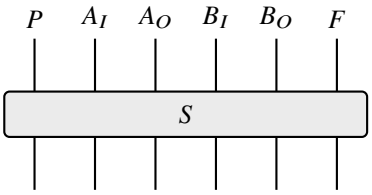
The map K in (a) is a matrix with rows "out" and columns "in"; if it is unitary it is full rank, and the cut below it crosses its "in" wire. Bending that wire upward with the identity state $|I\rangle\rangle = \sum_k |k\rangle|k\rangle$ gives the vector (b): the *same numbers* K_{mn}, now indexing a single vector on in $\otimes$ out. The Choi matrix (c) is that vector times its dual. As an operator on in $\otimes$ out it has two wires in and two out, like (d), but between the ket half and the bra half there is a gap that no wire crosses: that is what rank one means. Nothing about unitarity was used; (c) is rank one for any K.

5. Where ρ enters, and why $K^\dagger K = 1$

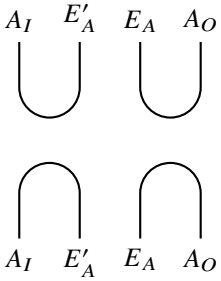


Multiplying ρ by K and then $K^\dagger$ and closing the wire is $\mathrm{Tr}(K^\dagger K\rho)$. Trace preservation of $\rho \mapsto K\rho K^\dagger$ demands that this equal $\mathrm{Tr}\,\rho$ for *every* ρ. The state is a separate box plugged into the open wire; it is not inside K. Since the equality holds for all ρ, the box ρ can be removed from the statement, leaving the operator identity $K^\dagger K = 1$ on the right. This uses only normalization; unitarity would add $KK^\dagger = 1$, i.e. K square and invertible.

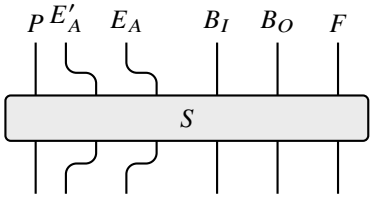
6. A pure process, and the swap argument



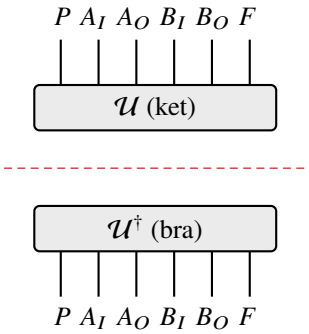
(a) S: an operator on all six wires



(b) the swap's Choi matrix $|R_A\rangle\rangle\langle\langle R_A|$: only identity wires



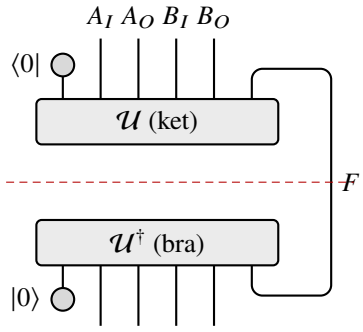
(c) S contracted with (b): the A_I, A_O wires are merely renamed E'_A, E_A



(d) purity $\Rightarrow S = |\mathcal{U}\rangle\rangle\langle\langle\mathcal{U}|$: no wire crosses the cut

(a) A process with past and future is an operator on six wires. Its Born rule contracts P with a past state, the four laboratory wires with $M_a \otimes N_b$, and F with a measurement on the future. (b) The local operation "swap A_I out to E'_A, E_A in to A_O" has a Choi matrix made of nothing but identity wires. (c) Contracting S with it therefore does nothing except carry the A_I wires to the E'_A position and the A_O wires to the E_A position: the result is S itself, now read as an operator with wires P, E_A, E_B and F, E'_A, E'_B, i.e. as the Choi matrix of a map $\mathcal{U}\colon P \otimes A_O \otimes B_O \to A_I \otimes B_I \otimes F$. Purity says this map is a unitary. Picture 4 then says that S is a ket over a bra with nothing between them, (d). Picture 5, with the "every ρ" running over states of $P \otimes E_A \otimes E_B$, says $\mathcal{U}^\dagger\mathcal{U} = 1$.

7. From S back to W: plugging in $|0\rangle$ and tracing out F



$W = \mathrm{Tr}_F[\langle 0|_P S|0\rangle_P]$: the cut now crosses the F wire

S contains no state on P; the state is supplied from outside, as the two dots $\langle 0|$ and $|0\rangle$ attached to the open P wires (one on the ket side, one on the bra side, because $W = \langle 0|S|0\rangle$ contracts both). Discarding F is a wire looped from the ket's F to the bra's F. What is left has only the four laboratory wires open: it is a bipartite process matrix W as in picture 2. And the horizontal cut that crossed nothing in picture 6(d) now crosses the F loop, so W can have rank up to $\dim F$. This is how a rank-one S produces a mixed W, and it is the step in the purification theorem where coherence between causal orders is lost.